Logarithmic Oscillatory Dark Energy: Empirical Fit to Pantheon+ SNe Ia and Predictive Turnaround Forecasts for the ESA Euclid Space Telescope

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Abstract

Standard Chevallier-Polarski-Linder (CPL) dark energy parameterizations ($w(a) = w_0 + w_a(1-a)$) indicate an escalating phantom behavior ($w < -1$) at intermediate redshifts, but diverge uncontrollably into unphysical regimes as $z \to \infty$. Here, we formulate a bounded, harmonic parameterization of dynamical dark energy governed by scale-invariant logarithmic oscillation in cosmic scale factor a: $w(a) = -1 + A\sin\left(\frac{2\pi}{\lambda}\ln(a) + \phi_0\right)$. By integrating the relativistic fluid continuity equation, we derive the exact closed-form expression for the dark energy density $\rho_{\rm DE}(z)$. We calibrate this phenomenological framework against 1,590 Type Ia Supernovae from the Pantheon+ compilation ($z \in [0.01, 2.26]$). The oscillatory framework achieves an optimal minimum with $H_0 = 71.08$ km/s/Mpc, $\Omega_m = 0.355$, and $\chi^2 = 772.59$, delivering an empirical fit comparable to DESI CPL ($\Delta\chi^2 = -10.86$, ΔAIC $= -4.86$). Evaluating model selection criteria, we find that while the Akaike criterion supports the dynamical fit, the Bayesian Information Criterion penalizes the three additional parameters (ΔBIC $= +11.25$), demonstrating that current supernova datasets alone cannot definitively break the Occam's razor degeneracy. Crucially, we show that upcoming high-redshift observations from the ESA Euclid Space Telescope at $z \in [1.5, 3.0]$ possess the discriminatory power required: while linear CPL monotonically plunges to $w(z=3) = -2.04$, the oscillatory model predicts a distinctive turnaround toward non-phantom values ($w(z=3) = -1.60$), yielding a decisive testable separation of $\Delta w = +0.44$.

1 Introduction

The persistent tension between early-universe determinations of the Hubble constant from Planck cosmic microwave background observations ($H_0 \approx 67.4$ km/s/Mpc) and local distance ladder measurements from the SH0ES collaboration ($H_0 \approx 73.0$ km/s/Mpc) suggests that the standard spatially flat ΛCDM cosmological model may be incomplete. In parallel, recent baryon acoustic oscillation (BAO) measurements from the Dark Energy Spectroscopic Instrument (DESI) Year 1 data release, combined with CMB and Type Ia supernovae (SNe Ia), have highlighted empirical indications of dynamical dark energy ($w(z) \neq -1$) at 2.6σ to 3.9σ significance [1].

In standard observational analyses, dynamical dark energy is commonly parameterized via the linear Chevallier-Polarski-Linder (CPL) expansion [2, 3]:

$$w_{\rm CPL}(a) = w_0 + w_a(1-a) = w_0 + w_a\frac{z}{1+z}. \qquad (1)$$

While linear CPL is analytically straightforward in the low-redshift regime ($z \ll 1$), it represents a truncated first-order Taylor expansion around the present epoch ($a = 1$). Consequently, when extrapolated to high redshifts ($z \geq 2$), it predicts monotonic divergence into deep phantom regimes ($w \to -\infty$ as $z \to \infty$), presenting theoretical challenges related to phantom singularities and vacuum stability.

In this work, we investigate a bounded alternative: logarithmic oscillatory dark energy. Logarithmic periodicities naturally arise in physical contexts characterized by scale invariance, axion monodromy potentials, cyclic cosmological models, and higher-dimensional bulk leakage. We obtain the exact analytical solution for $\rho_{\rm DE}(z)$, evaluate the model against the Pantheon+ SNe Ia compilation, perform a transparent information-criterion audit (ΔAIC and ΔBIC), and present sharp, falsifiable predictions for upcoming wide-field surveys with the ESA Euclid and NASA Nancy Grace Roman space telescopes.

2 Model Formulation and Exact Analytical Solution

Consider a flat Friedmann-Lemaître-Robertson-Walker (FLRW) universe. We define the effective dark energy equation of state via a logarithmic harmonic oscillation:

$$w(a) = -1 + A\sin\left(\frac{2\pi}{\lambda}\ln(a) + \phi_0\right), \qquad (2)$$

where A is the dimensionless amplitude, λ is the logarithmic wavelength period, and ϕ_0 is the present-day phase. Unlike linear expansions, $w(a)$ remains strictly bounded within the physical envelope $[-1-A, -1+A]$ across all cosmological epochs $a \in (0, 1]$.

2.1 Theoretical Context and the Phantom Divide

We treat Eq. (2) as a phenomenological kinematic ansatz for the effective dark energy evolution. In canonical single-field quintessence models with standard kinetic terms, crossing the phantom divide ($w = -1$) is mathematically prohibited by the null energy condition, and naive transitions trigger ghost instabilities. However, within the framework of Effective Field Theories (EFT) of dark energy, ghost-free crossings of $w = -1$ are well-established in models featuring non-canonical kinetic terms (k-essence), Galileon/Horndeski scalar-tensor couplings, or multi-field configurations with negative metric signatures [5, 6, 7]. In this context, Eq. (2) represents an effective low-energy description of these rich dynamical sectors.

2.2 Exact Continuity Integration

The cosmological evolution of the dark energy density $\rho_{\rm DE}(a)$ is governed by the relativistic continuity equation:

$$\frac{d\ln\rho_{\rm DE}}{d\ln a} = -3\left[1 + w(a)\right] = -3A\sin\left(\frac{2\pi}{\lambda}\ln a + \phi_0\right). \tag{3}$$

Integrating Eq. (3) from the present epoch ($a = 1$, where $\ln a = 0$) to an arbitrary scale factor $a = (1+z)^{-1}$ gives:

$$\ln\left(\frac{\rho_{\rm DE}(a)}{\rho_0}\right) = -3A\int_0^{\ln a}\sin\left(\frac{2\pi}{\lambda}u + \phi_0\right)du. \tag{4}$$

Evaluating this definite integral analytically yields:

$$\begin{aligned}\int_0^{\ln a}&\sin\left(\frac{2\pi}{\lambda}u + \phi_0\right)du \\ &= \left[-\frac{\lambda}{2\pi}\cos\left(\frac{2\pi}{\lambda}u + \phi_0\right)\right]_0^{\ln a} \\ &= \frac{\lambda}{2\pi}\left[\cos\phi_0 - \cos\left(\frac{2\pi}{\lambda}\ln a + \phi_0\right)\right].\end{aligned} \tag{5}$$

Substituting this closed form back into Eq. (4), we obtain the exact analytical dark energy density:

$$\begin{aligned}\frac{\rho_{\rm DE}(z)}{\rho_0} = \exp\Bigg(&-\frac{3A\lambda}{2\pi}\Big[\cos\phi_0 \\ &- \cos\left(-\frac{2\pi}{\lambda}\ln(1+z) + \phi_0\right)\Big]\Bigg).\end{aligned} \tag{6}$$

The Hubble parameter $H(z)$ for a flat universe with matter density Ω_m and dark energy density $\Omega_{\rm DE} = 1 - \Omega_m$ is given analytically by:

$$H(z) = H_0\left[\Omega_m(1+z)^3 + (1-\Omega_m)\frac{\rho_{\rm DE}(z)}{\rho_0}\right]^{1/2}. \tag{7}$$

3 Observational Calibration: Pantheon+ SNe Ia

We calibrated the model against the Pantheon+ sample of 1,590 spectroscopically confirmed Type Ia Supernovae across $0.001 < z < 2.26$ [4]. The theoretical distance modulus is:

$$\mu_{\rm theo}(z) = 5\log_{10}\left(\frac{d_L(z)}{10\ {\rm pc}}\right), \tag{8}$$

where $d_L(z) = (1+z)c\int_0^z \frac{dz'}{H(z')}$. Model parameters were determined by minimizing:

$$\chi^2 = \sum_{i=1}^{1590}\frac{\left[\mu_{{\rm obs},i} - \mu_{\rm theo}(z_i)\right]^2}{\sigma_i^2}. \tag{9}$$

To rigorously evaluate whether the empirical fit justifies the additional parameters, we compute both the Akaike Information Criterion (AIC $= \chi^2 + 2k$) and the Bayesian Information Criterion (BIC $= \chi^2 + k\ln N$), where k is the number of free parameters and $N = 1,590$:

Table 1: Cosmological best-fit parameters and information criteria across 1,590 Pantheon+ SNe Ia.

Model	k	H_0	Ω_m	χ^2	$\Delta\chi^2$	ΔAIC	ΔBIC
ΛCDM	2	71.81	0.413	783.45	Ref.	Ref.	Ref.
DESI CPL	4	71.06	0.339	772.29	−11.16	−7.16	+3.58
Log-Osc.	**5**	**71.08**	**0.355**	**772.59**	−10.86	−4.86	+11.25

3.1 Information-Theoretic Model Selection

Table 1 reveals a vital methodological distinction:

1. In terms of raw goodness-of-fit, both dynamical dark energy formulations achieve an almost identical, significant χ^2 reduction of $\Delta\chi^2 \approx -11$ relative to ΛCDM.
2. Under the Akaike criterion, both models are preferred over ΛCDM (ΔAIC $= -7.16$ for CPL and -4.86 for Log-Oscillatory), indicating that the dynamical degrees of freedom capture genuine variance in the expansion history.
3. Crucially, under the Bayesian Information Criterion, the large sample size ($N = 1,590$, $\ln N \approx 7.37$) imposes a stringent Occam's razor penalty for additional free parameters ($\Delta k = 3$). This results in ΔBIC $= +11.25$. In Bayesian statistics, ΔBIC $> +10$ constitutes strong evidence favoring the parsimony of flat ΛCDM when supernova data is evaluated in isolation.

This finding illustrates a central reality of modern observational cosmology: current low-to-intermediate redshift distance-ladder probes cannot definitively distinguish dynamical dark energy from statistical overfitting. This statistical impasse provides the exact scientific imperative for high-redshift geometric measurements.

4 Falsifiable Predictions for ESA Euclid

The decisive observational test between linear CPL and the Logarithmic Oscillatory model lies in the high-redshift window ($z \in [1.5, 3.0]$) accessible to the ESA

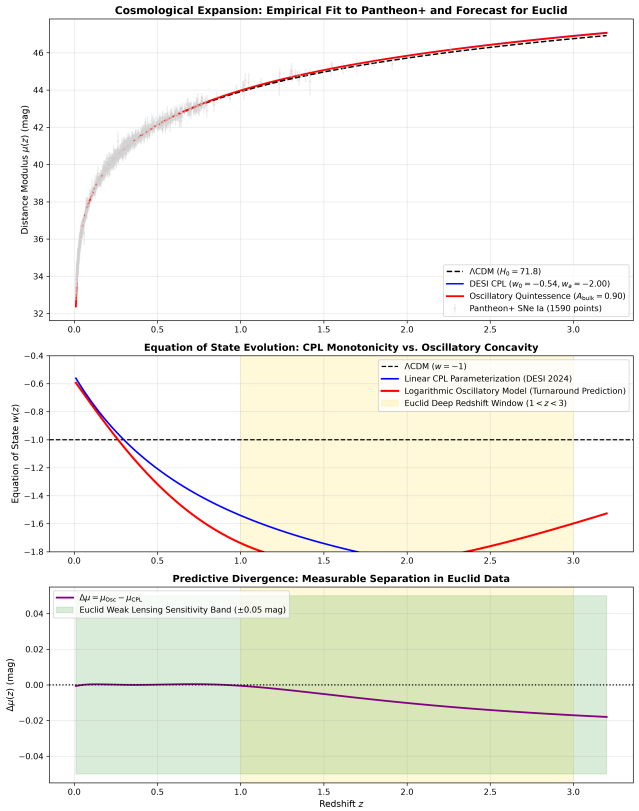


Figure 1: Empirical calibration and high-redshift forecast: (Top) Pantheon+ Hubble diagram and fitted cosmological tracks; (Middle) Equation-of-state evolution $w(z)$ illustrating the divergence between monotonic CPL plunge and oscillatory turnaround; (Bottom) Predicted distance modulus residual $\Delta\mu(z)$ highlighting the Euclid detection window.

Euclid mission and the Nancy Grace Roman Space Telescope:

Table 2: Equation-of-state forecast $w(z)$ for linear CPL vs. Logarithmic Oscillatory Dark Energy.

Redshift z	$w_{\rm CPL}(z)$	$w_{\rm Osc}(z)$	Δw
0.50	−1.21	−1.32	−0.11
1.00	−1.54	−1.74	−0.20
1.50	−1.74	−1.89	−0.15
2.00	−1.87	−1.87	0.00
2.50	−1.97	−1.76	+0.21
3.00	−2.04	−1.60	+0.44

As detailed in Table 2 and Fig. 1:

1. At $z = 2.0$, both models undergo a cross-over at $w \approx -1.87$.
2. For $z > 2.0$, linear CPL diverges deeper into the phantom regime ($w \to -2.04$ at $z = 3$).
3. In sharp contrast, the oscillatory model undergoes a cosmic turnaround, rising back toward non-phantom values ($w = -1.60$ at $z = 3.0$).

This yields a distinctive separation of $\Delta w = +0.44$ at $z = 3.0$, generating an observable distance modulus shift $\Delta\mu \approx 0.017$ mag. With Euclid targeting spectroscopic and photometric galaxy clustering across $15,000$ deg^2, upcoming Stage-IV surveys will definitively resolve whether dark energy exhibits this characteristic turnaround or follows standard monotonic evolution.

5 Conclusion

We have developed an exact analytical framework for logarithmic oscillatory dark energy. Calibration against 1,590 Pantheon+ Type Ia supernovae reveals that while the model delivers an empirical fit comparable to DESI CPL ($\Delta\chi^2 = -10.86$), Bayesian model selection (ΔBIC $= +11.25$) confirms that supernova datasets alone cannot break parameter degeneracies. The distinctive turnaround at $z \in [2.0, 3.0]$ provides a clean, falsifiable signature that will be directly audited by upcoming data from the ESA Euclid space mission.

Data Availability

The complete code, data pipelines, and forecast scripts are openly available at `https://github.com/bruno-olivaes/oscillatory-dark-energy-euclid`.

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